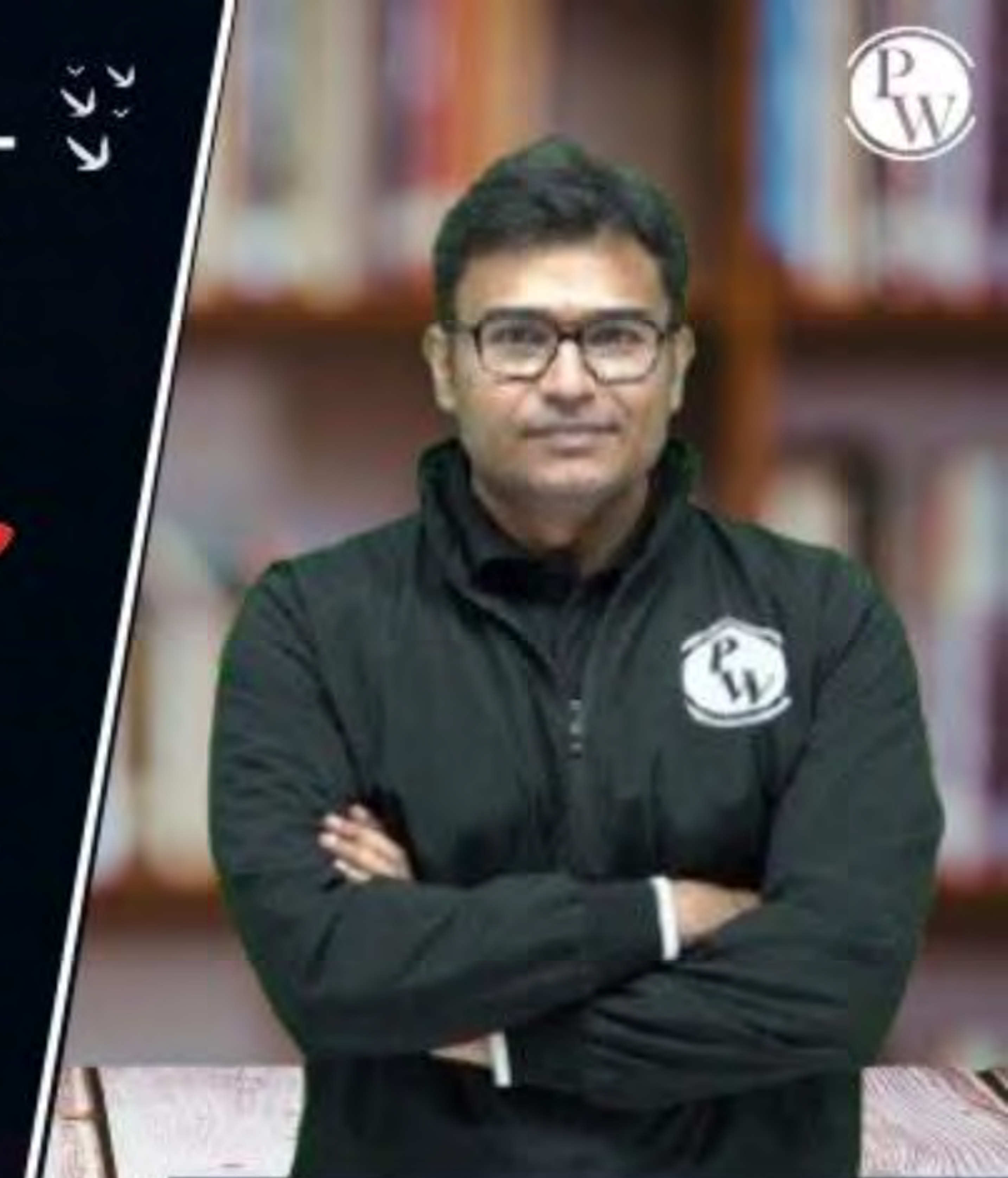
Computer Science & IT

Theory of Computation

Turing Machines

Lecture No. 1



By- Ravindra Sir

Recap of Previous Lecture

Topic CFG

Topic

Topic

Topics to be Covered

Topic TM

Topic

Topic

→ If for a language 'L', there is an ambiguous grammar then L is

a) ambi b) unambi c) cant say

→ If for a language L, there is unambiguous grammar the L is

a) ambi ~~b)~~ unambi c) can't say

$$L \left\{ \left. \begin{matrix} G_1 \\ G_2 \\ G_3 \\ \vdots \\ G_n \end{matrix} \right\} \text{ambi} \right.$$

$L \downarrow$ ambi

L — G (unambi)

$L \downarrow$ unambi

Rec, RE Languages and TM

RE $\rightarrow$ TM

Rec — Halting TM

CSL — LBA — linear bounded automata

CFL — PDA

DCFL — DPDA

RL — FA

TM :-

1. TM theory
2. $M \rightarrow L(M)$
3. $L(M) \rightarrow M$
4. TM as a transducer

$$f(x, y) \rightarrow x + y$$

5. Church turing thesis
   → TM is as powerful as logic
6. Variations of TM ✓
   No variation has more power than Deterministic TM ✓

Rec and RE languages :-

1 Definition

1. Rec
2. RE but not Rec

$1+2 \to$ RE

3. non RE

$2^{\Sigma^*} \to$ Set of all languages

2. Properties:-

Enumeration procedure — listing all the strings of a language

membership procedure — Both members and non members

Proper ordering

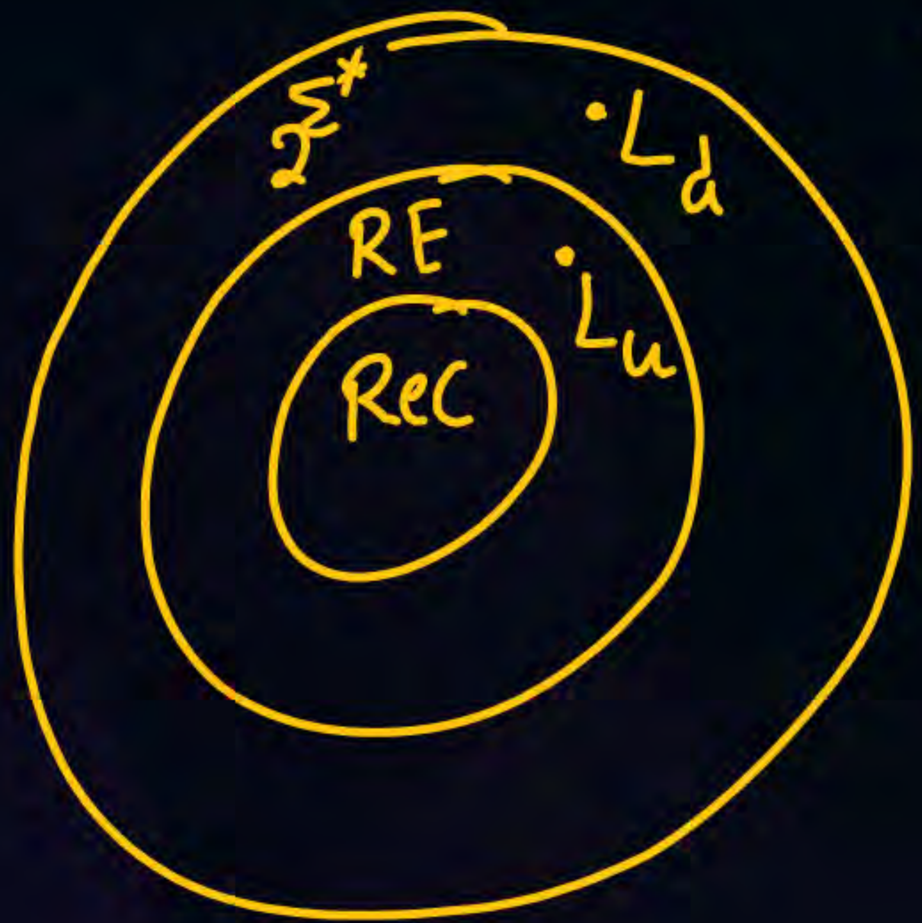
$\Sigma = \{a, b\}$

$\{\epsilon, a, b, aa, ab, ba, bb \ldots\}$

closure

L and $\bar{L}$ theorem

3)



$L_u \rightarrow$ universal language of TM

$L_d \rightarrow$ diagonalisation of TM

4) undecidable problems in TM

27/40 → undecidable

6 more undecidable problems

5) Rice's theorem

TM theory :-

$M(Q, \Sigma, \Gamma, \delta, q_0, B, F)$

- Q → States
- Σ → i/p alphabet
- Γ → Tape alphabet
- δ → Transition function
- q_0 → initial state
- B → Blank symbol $\square$
- F → Set of final states

$\Sigma : \{a, b\} \subseteq \Gamma = \{a, b, \underline{\underline{X, Y}}, B\}$

$\cdots \; \underbrace{B \mid B \mid B \mid a \mid a \mid b \mid b}_{} \mid B \mid B \mid B \; \cdots$

DTM : $\delta : Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$

NTM : $\delta : Q \times \Gamma \rightarrow 2^{Q \times \Gamma \times \{L, R\}}$

$$\delta(q_0, a) = \{(q_1, b, R), (q_2, b, L)\}$$

acceptance:
Halt final state

Rejection
a) Dead config (halt on non final)
b) Hanging ✓

End ✓

→ There is an algo to convent NTM to DTM

↓

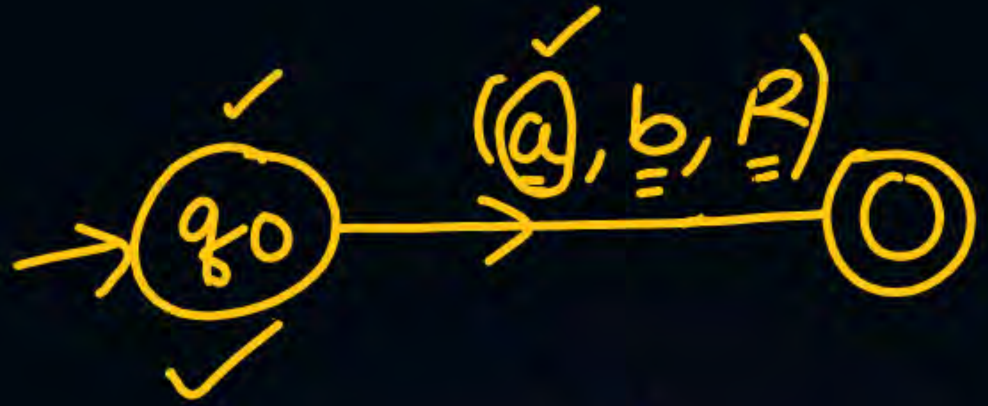
decidable

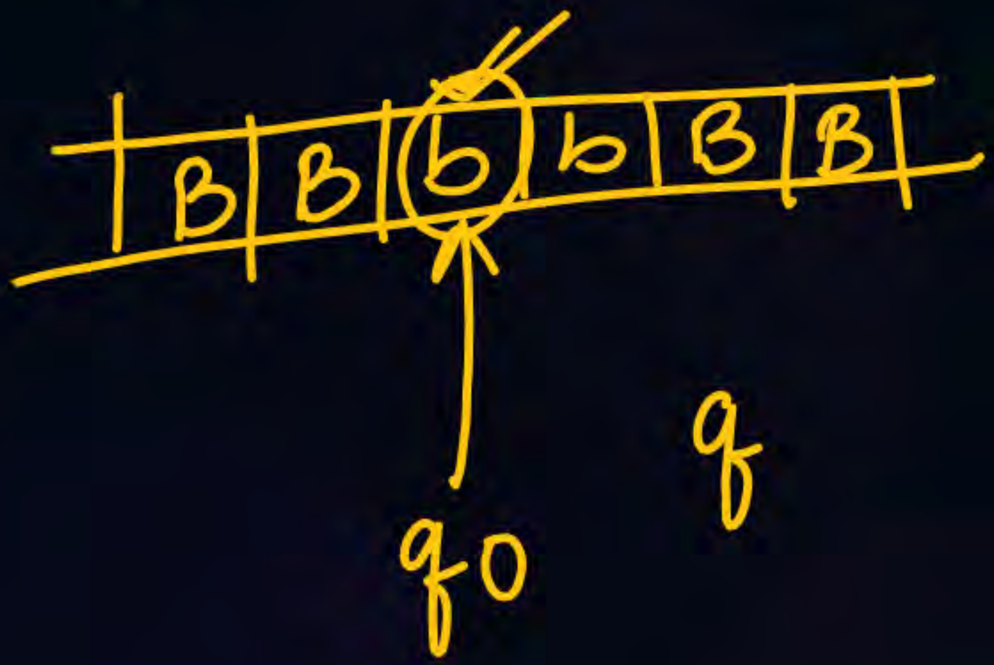
Power of $\underline{\underline{\text{NTM}}}$ = power of $\underline{\underline{\text{DTM}}}$

$$\left.\begin{array}{l} DFA = NFA \\ NPDA > DPDA \\ DTM = NTM \end{array}\right\}$$

Tm
- Halt
  - halt in final state → accepted
  - halt in non final state → Reject
- Hang → Reject

Tm always stops in dead configuration


q0
(a, b, R)


B B b b B B
q0
q

Rec → Halting TM → always halts on any i/p

RE → TM → may halt or may not halt

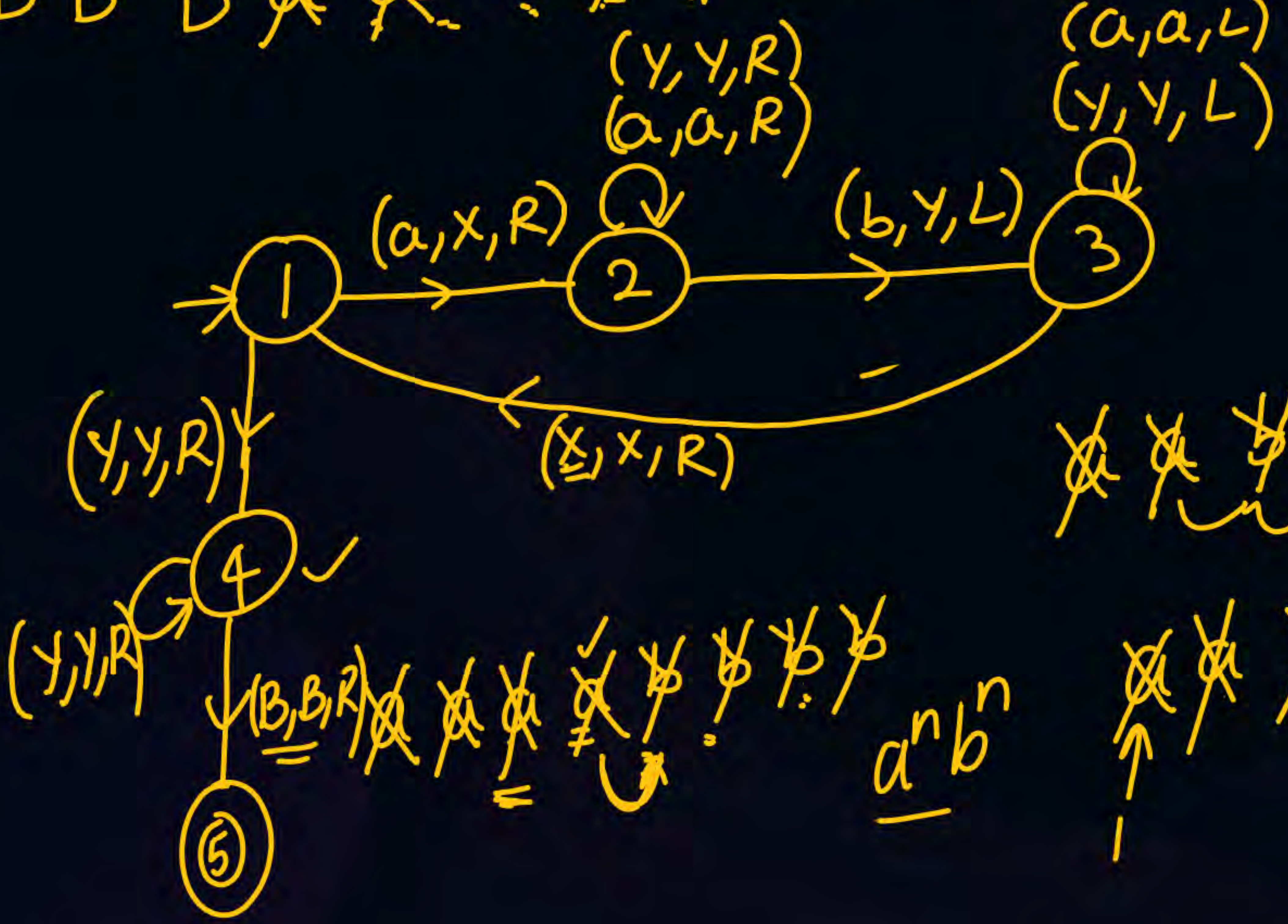
Sand bags ✓ ✓ cement bags ✓

$a^n b^n / n \geq 1$

$\underline{\cancel{a}\,\cancel{a}\,\cancel{a}}\ \underline{\cancel{b}\,\cancel{b}\,\cancel{b}}$

5 min ✓

....BB B $\cancel{a}$ $\cancel{a}$ a $\cancel{b}$ $\cancel{b}$ b B B B.....



B $\cancel{a}$ $\cancel{b}$ b B

4

$\cancel{a}$ $\cancel{a}$ $\cancel{b}$ $\cancel{b}$ B

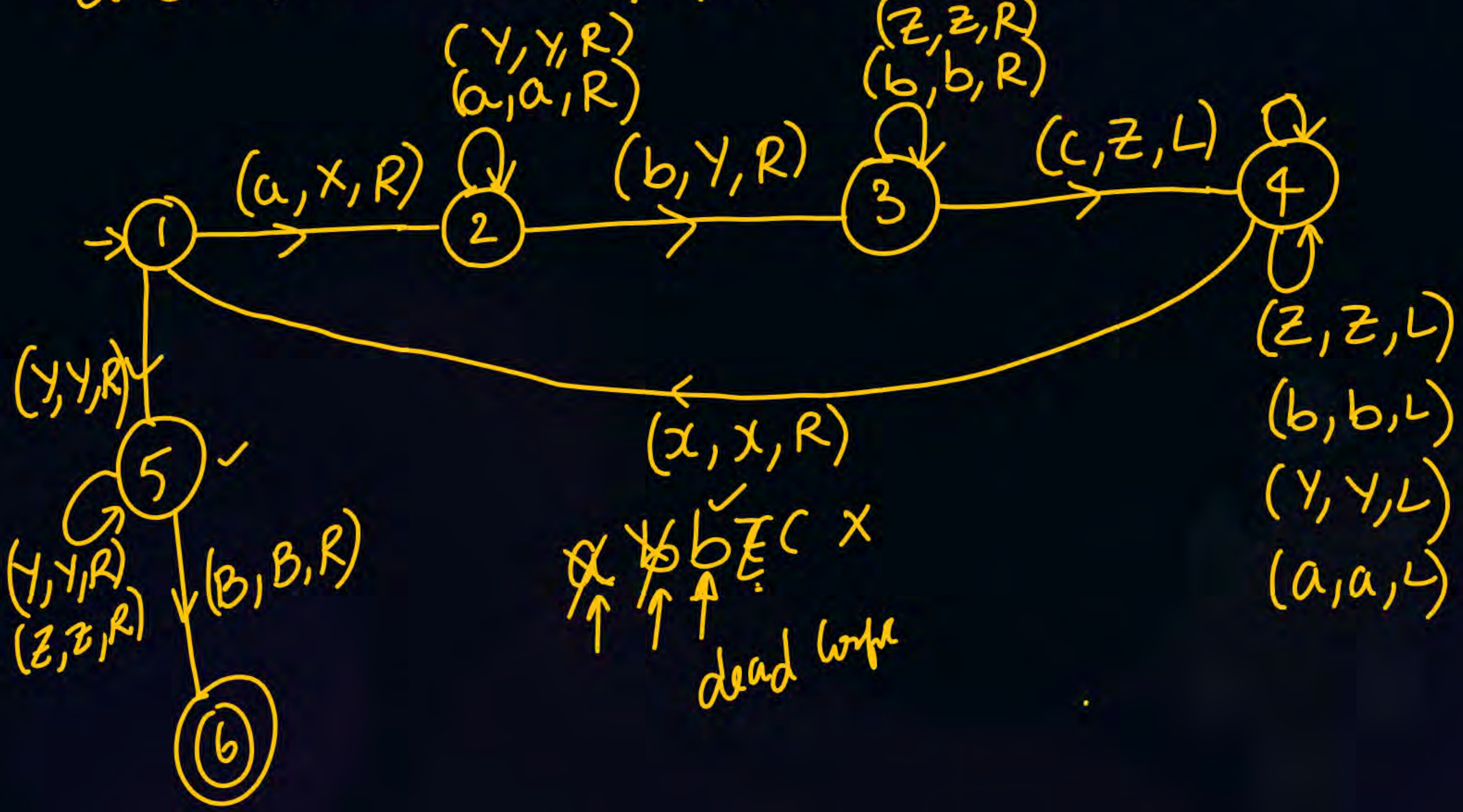
$\cancel{a}$ $\cancel{a}$ $\cancel{a}$ $\cancel{a}$ $\cancel{b}$ $\cancel{b}$ $\cancel{b}$ $\cancel{b}$

$a^n b^n$

$\cancel{a}$ $\cancel{a}$ $\cancel{b}$ B ✓

1 2

$a^n b^n c^n / n \geq 1$ B B B ~~X~~ ~~X~~ ~~X~~ a ~~Y~~ ~~Y~~ ~~Y~~ b Z Z Z c B B B



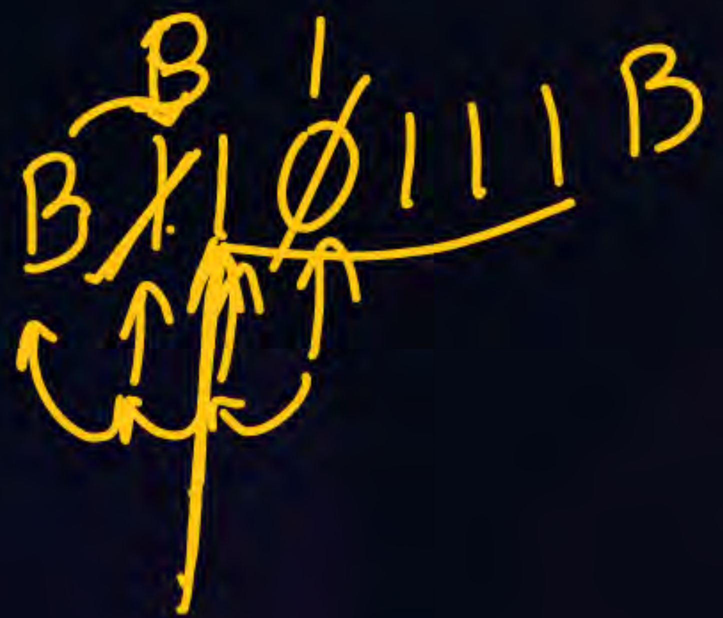
TM as a transducer: I/p → O/p

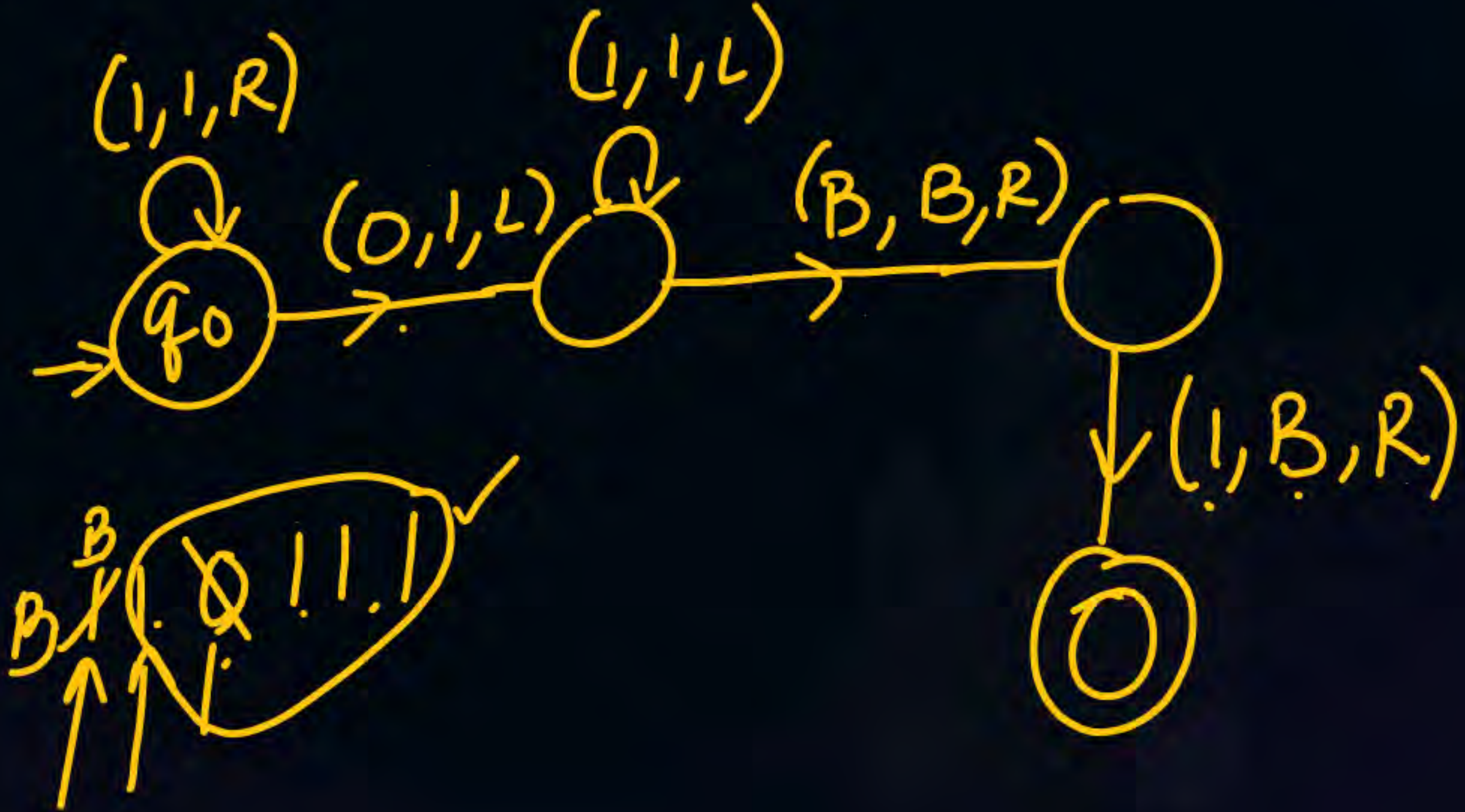
$f(x, y) = x + y$

$2 + 3$

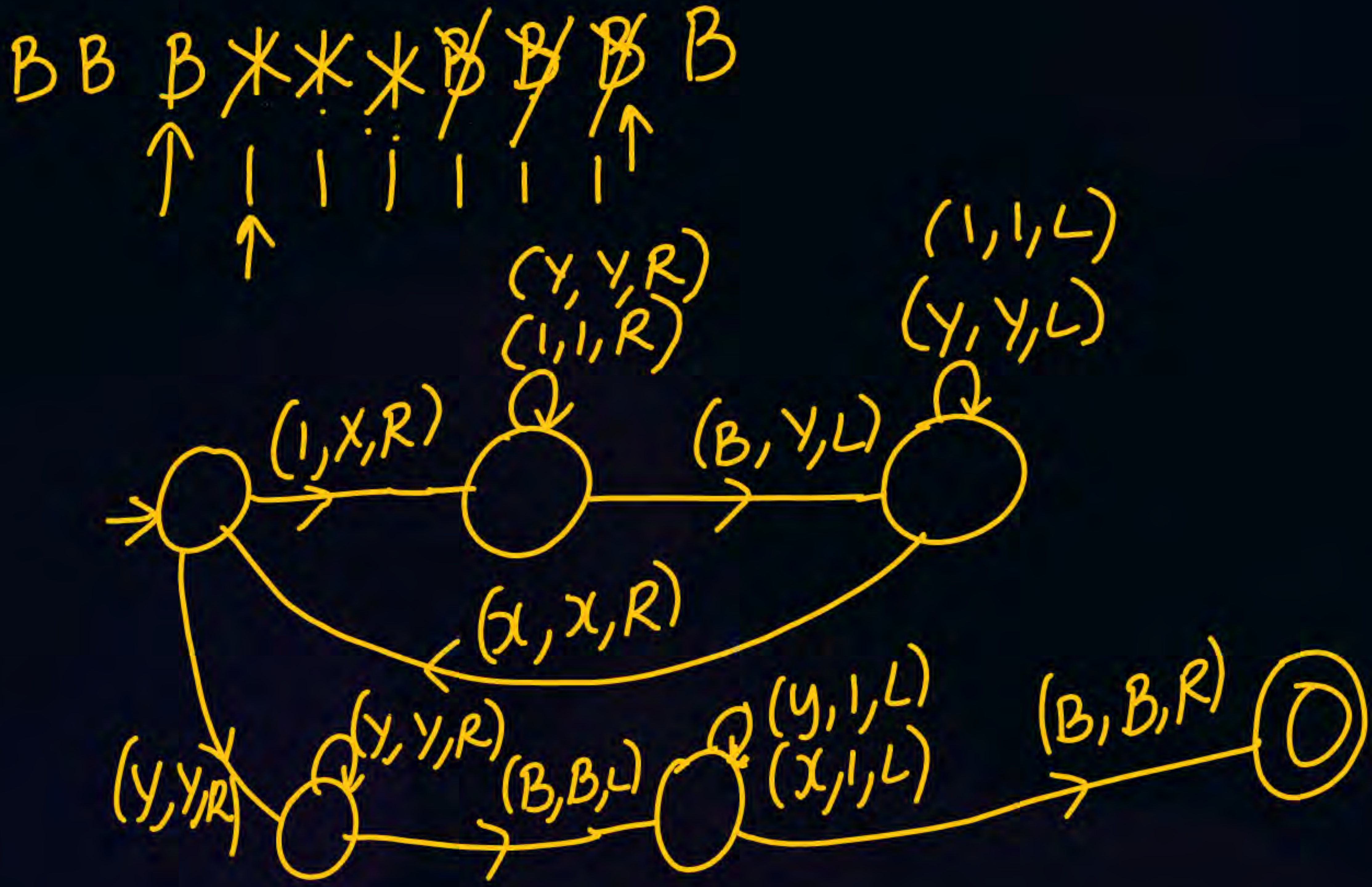
B B 1 1 0 1 1 1 B B

B B 1 1 1 1 1 B B





$f(x) = 2x \qquad 111 \rightarrow 111111$



Church turing thesis :-

Every logically computable function is Turing computable.

Every logically recognisable function is Turing recognisable.

Turing recognisable $\rightarrow$ RE language

$\downarrow$

halts only on members of the language

Turing decidable $\rightarrow$ Rec language

$\downarrow$

halts on all strings

Variations of TM :-

TM $\rightarrow$ FA + R/W + R $\leftrightarrow$ L

Standard TM :-

1) Deterministic TM
2) FA + R/W Tape
3) R $\leftrightarrow$ L
4) Single tape (infinite)

Variations of TM :-

1. TM with stay option $\delta : Q \times \Gamma \rightarrow Q \times \Gamma \times \{\underline{\underline{L}}, \underline{\underline{R}}, \underline{\underline{S}}\}$

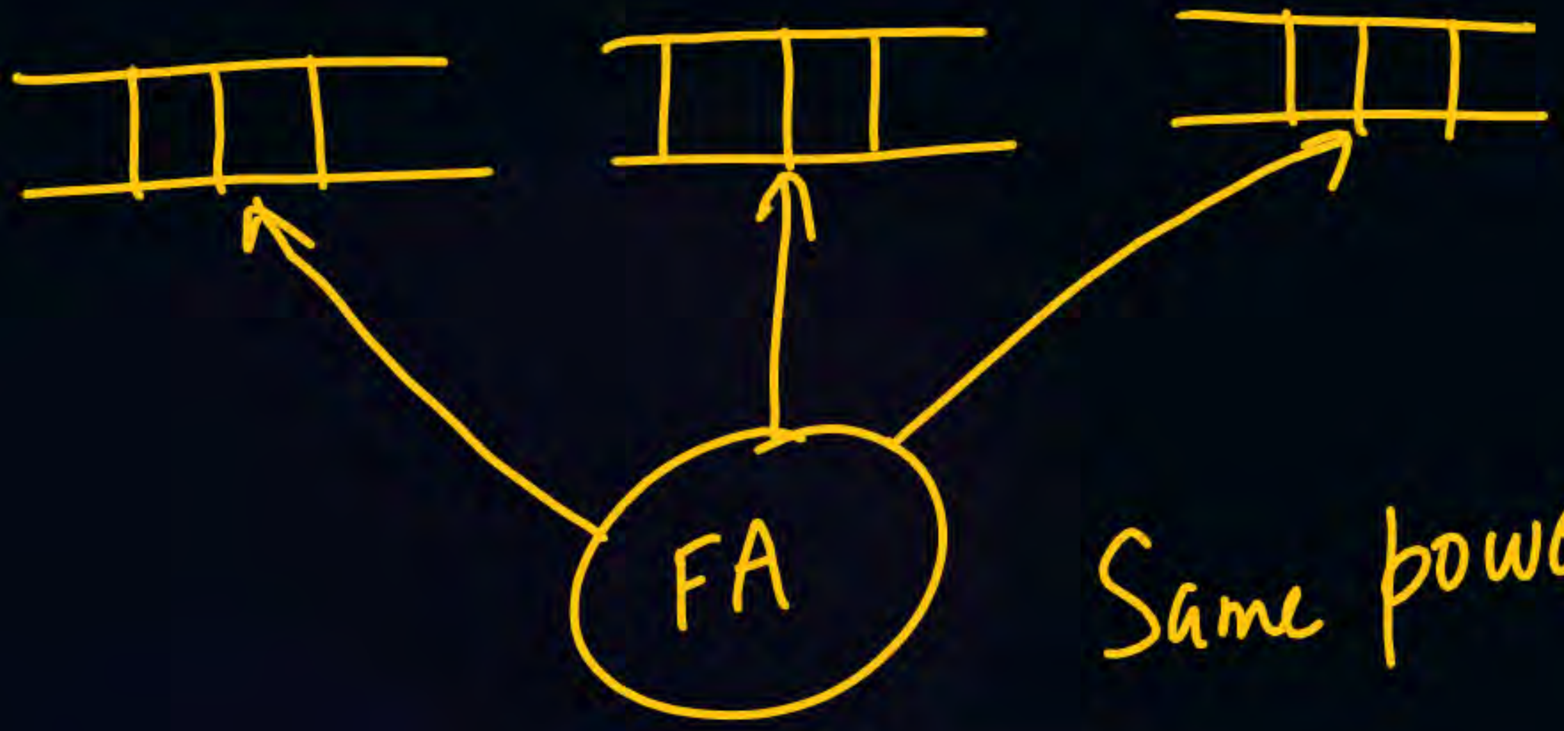
Same power as STM

2. TM with semi infinite tape

$\$abBBBB\cdots$

Same power as STM

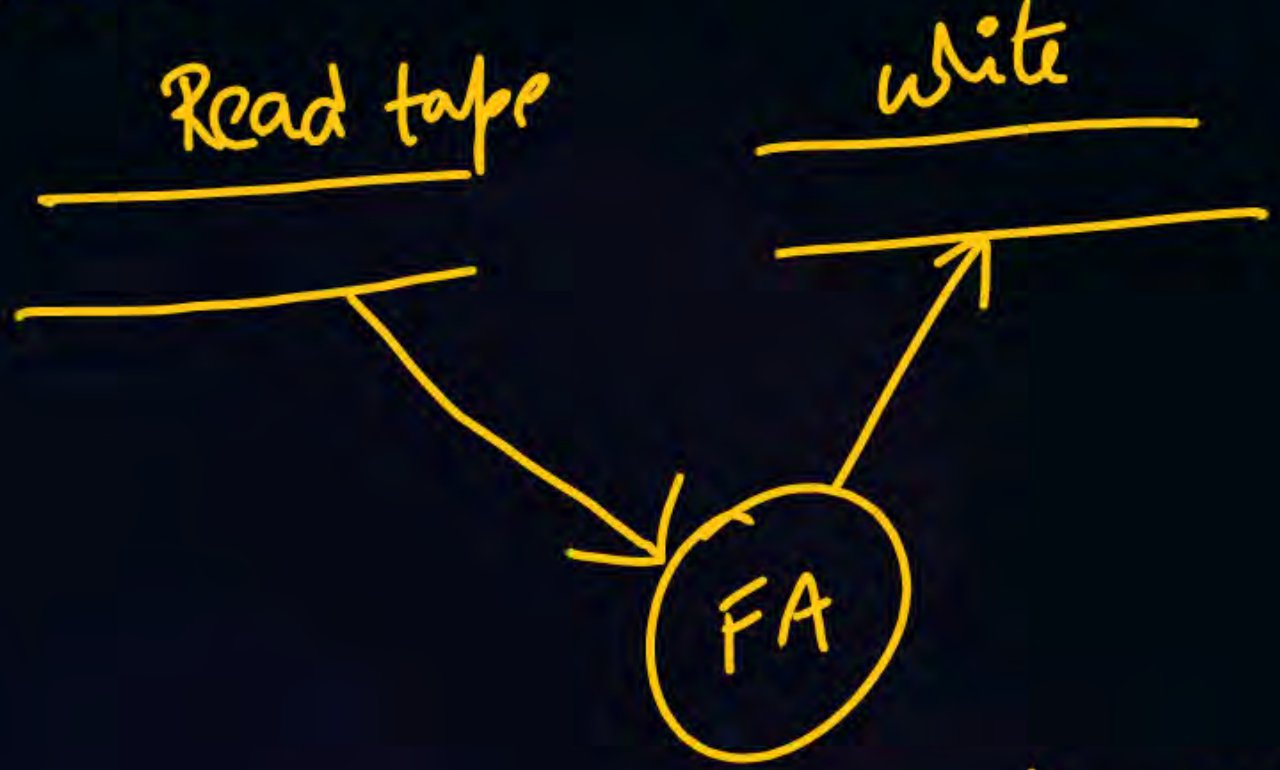
3) multi tape TM



Same power as STM

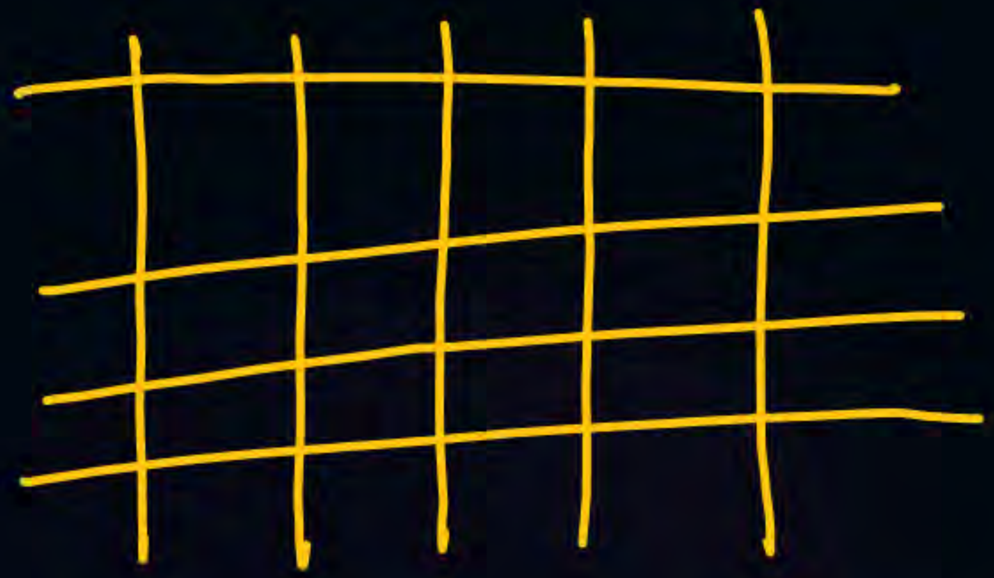
4) offline TM

Read and write tapes will be different



Same power as STM

5) multi dimensional TM



R ⟷ L   U ⟷ D

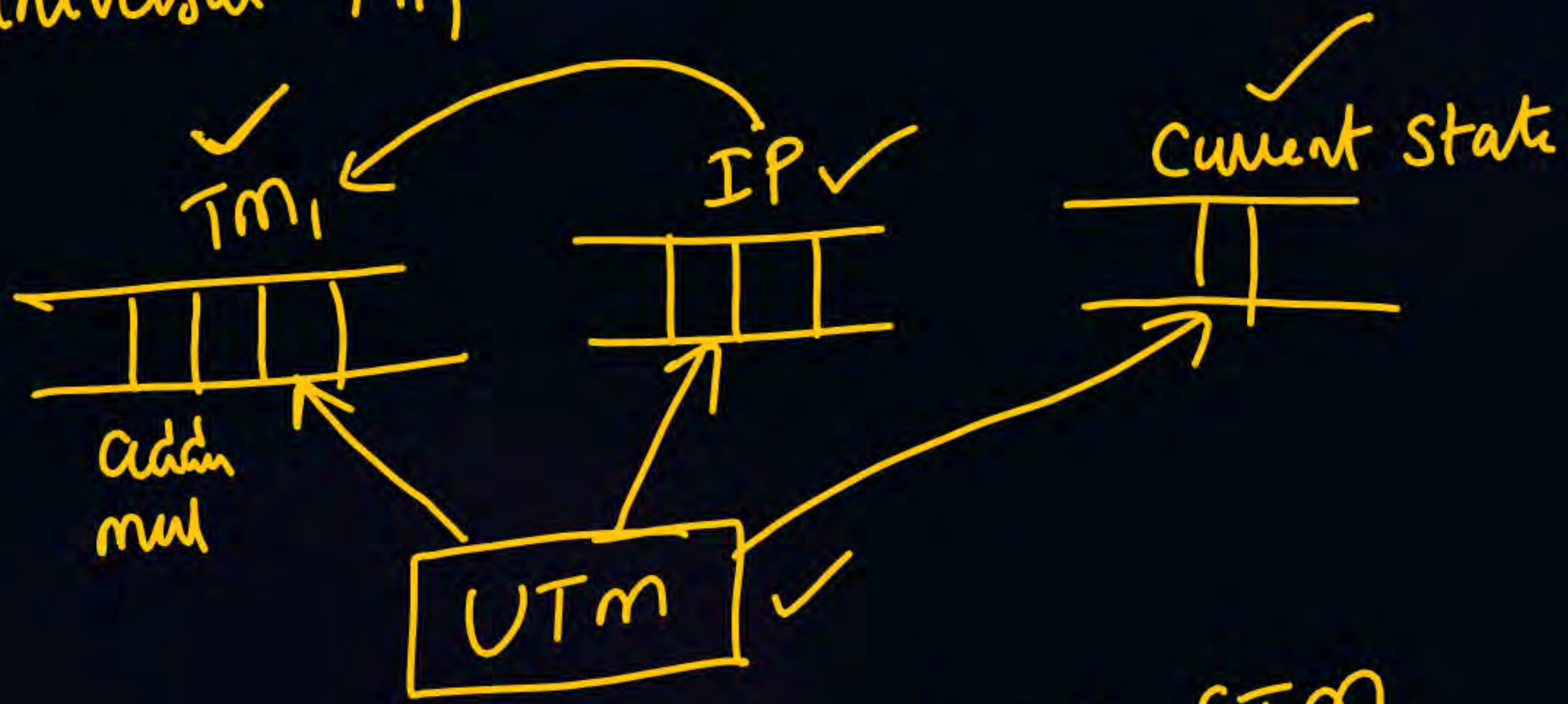
$$\delta : Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R, U, D\}$$

Same power STM

6) Non deterministic TM

Same power as STM

7) universal TM

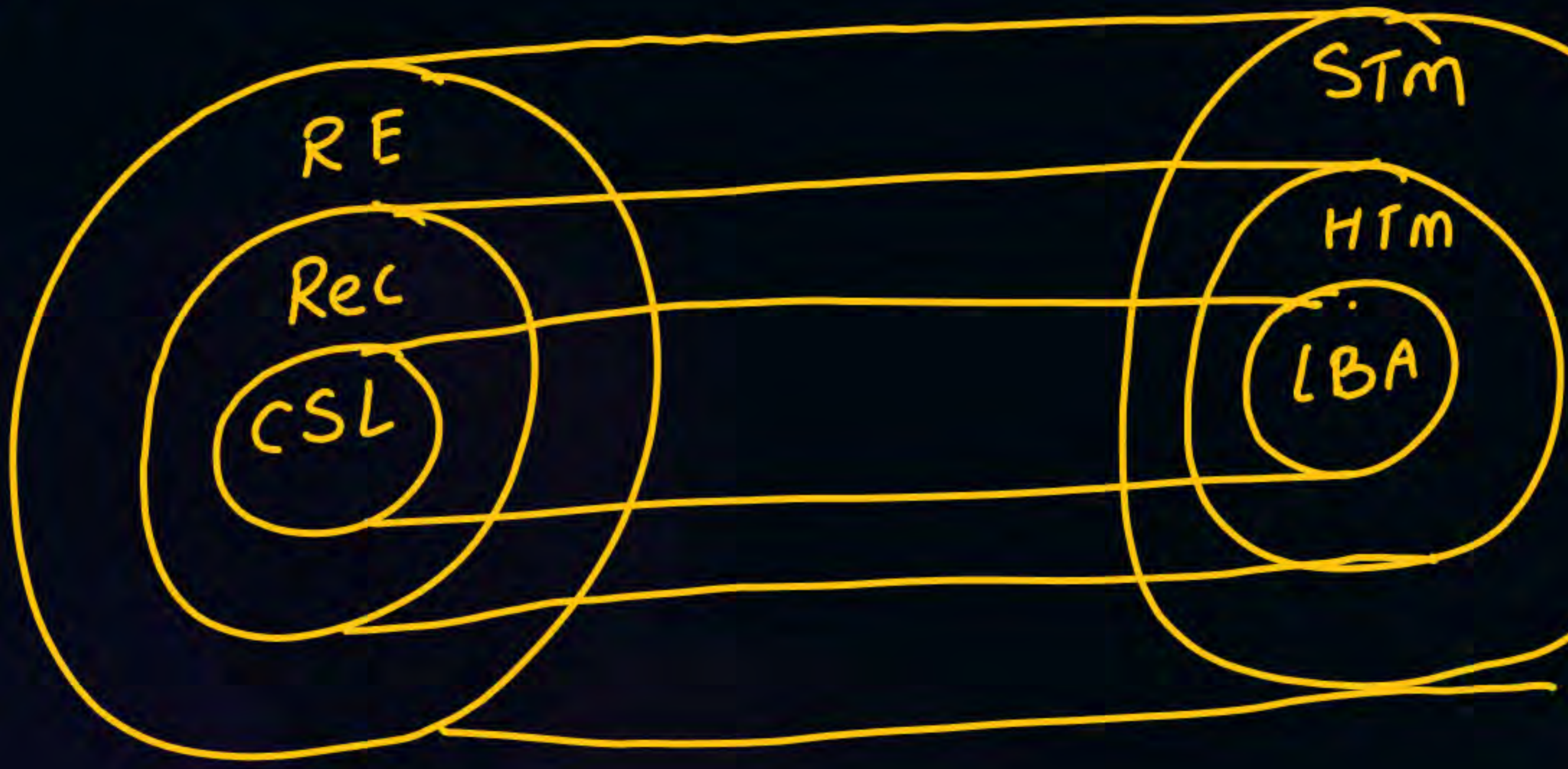


Same power as STM

4 more variations:-

1. Halting TM less powerful than Standard TM

2) LBA < HTM < STM



LBA $\rightarrow$ tape is not infinite

\$ a a b b \$

LBA can only use the space where i/p is present

3) Read only TM $\cong$ 2-way $\underset{=}{FA}$ $\cong$ FA

no writing

4) one way TM $FA + R/W + (R \cap L) \cong FA$

THANK - YOU